

Display disk space usage for all directories with hierarchical breakdown

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Shows disk space usage for directories up to 2 levels deep in the current directory tree, sorted by size in descending order with human-readable format, providing a hierarchical breakdown of disk consumption.

Language

bash

Category

disk-usage

Command

```
du -h --max-depth=2 . | sort -hr
```

Explanation

The command uses 'du -h' to calculate disk usage in human-readable format. The '--max-depth=2' flag limits output to directories up to 2 levels deep from the current directory, avoiding excessive detail while still showing the hierarchy. The '.' specifies the starting directory. The output is piped to 'sort -hr' to sort in descending order by human-readable sizes, making it easy to identify which directories consume the most space at each level.

Tags

du, disk-usage, directories, storage, hierarchical, sort, system-administration

Dependencies

coreutils

Arguments

1. **PATH** (Optional): Root directory to analyze (default: current directory)
Default: .
2. **DEPTH** (Optional): Maximum directory depth to traverse (default: 2)
Default: 2

Examples

1. `du -h --max-depth=2 . | sort -hr` - Show disk usage up to 2 levels deep from current directory
2. `du -h --max-depth=1 . | sort -hr` - Show only immediate subdirectories (1 level deep)
3. `du -h --max-depth=3 /home | sort -hr` - Show disk usage in /home up to 3 levels deep
4. `du -h --max-depth=2 . | sort -hr | head -20` - Show top 20 directories by size (2 levels deep)
5. `du -ah --max-depth=2 . | sort -hr | head -15` - Include files in the listing, show top 15 items

Output

```
1.2T    .
847G    ./data
512G    ./data/cache
256G    ./data/backups
244G    ./media
128G    ./media/videos
116G    ./media/archives
156G    ./documents
96G     ./documents/projects
60G     ./documents/reports
```

Notes

- The `--max-depth` flag controls recursion depth: 0 shows only the starting directory, 1 shows immediate children, etc.
- The `-h` flag makes output human-readable (K, M, G, T); use `-B 1M` for sizes in fixed block units

- The `-s` flag is not needed here as `du` summarizes by default
- Hidden directories (those starting with `.`) are included automatically
- Each line shows cumulative size for that directory and all its contents
- Symbolic links are not followed by default; add `-L` to follow symlinks
- On filesystems with many nested directories, increasing `--max-depth` may significantly increase runtime

Warnings

- Results may be incomplete if you lack read permissions on some directories
- On network filesystems (NFS, SMB), this command can be slow due to network latency
- Very deep directory hierarchies combined with high `--max-depth` values may cause excessive output
- Sparse files may report misleading sizes on some filesystems

See Also

- `disk-space-sort-largest-directories`

Status

reviewed

Safety

safe

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

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